A. Build a Valid Induction Plan

The five required pieces

An induction proof establishes a proposition $P(n)$ for every integer $n \geq n_0$ by giving:

1. a precise statement of $P(n)$ and its domain;
2. a verified **base case** $P(n_0)$;
3. an **induction hypothesis** for an arbitrary $k \geq n_0$;
4. an **induction step** deriving $P(k+1)$ from the hypothesis; and
5. a conclusion invoking induction.

The hypothesis is temporary: it may be used only while proving the next case.

Ordinary versus strong induction

Ordinary induction assumes only $P(k)$ and proves $P(k+1)$. **Strong induction** assumes every case

$$P(n_0), P(n_0+1), \ldots, P(k)$$

and proves $P(k+1)$. The two principles are logically equivalent; choose the form that naturally supplies the earlier cases needed in the step.

Prime and composite integers

A **prime** is an integer at least 2 whose only positive divisors are 1 and itself. A **composite** integer is an integer at least 2 that is not prime: it can be written ab with integers $a, b \geq 2$.

A planning template

Before calculating, write:

1. $P(n)$: the exact claim with the variable visible;
2. Base: substitute the first permitted value;
3. IH: "Fix arbitrary $k \geq n_0$ and assume ...";
4. Goal: write $P(k+1)$ explicitly;
5. Bridge: identify the expression where the IH will enter.

For strong induction, state the entire range of assumed cases.

A.1: prime factorization setup

Problem (Core A.1). Consider the claim that every integer $n \geq 2$ is a product of one or more primes. State (a) the base case, (b) the strong induction hypothesis for an arbitrary $k \geq 2$, (c) what must be proved for $k+1$, and (d) the two cases considered in the induction step. Do not complete the proof.

Worked solution. Write $P(n)$ for the stated factorization claim.

Base. $P(2)$ holds because 2 is prime.

Strong IH. Fix arbitrary $k \geq 2$ and assume $P(j)$ holds for every integer j with $2 \leq j \leq k$.

Goal. Prove $P(k+1)$: that $k+1$ is a product of primes.

Cases. If $k+1$ is prime, it is already a product of one prime. If it is composite, write $k+1 = ab$ with $2 \leq a, b \leq k$; the strong IH applies to both factors. These are the two cases the short Core problem asks you to identify; do not assume a factorization before entering the composite case.

Answer. Base $P(2)$; assume $P(j)$ for all integers $2 \leq j \leq k$; prove $P(k+1)$; split into $k+1$ prime or composite. No full proof is requested.

A similar ordinary-induction plan

Problem (similar example). For every integer $n \geq 1$, let $P(n)$ be $1+3+\cdots+(2n-1) = n^2$. State the base case, ordinary induction hypothesis, and next-case goal. Identify the next term and show the algebra that connects the hypothesis to the goal.

Worked solution. 1. State and start. For $P(n)$: $1+3+\cdots+(2n-1) = n^2$, use base $n = 1$. **2. Assume the k case:**

$$1+3+\cdots+(2k-1) = k^2.$$

3. Write the target. The next new term is $2(k+1)-1 = 2k+1$. **4. Build the bridge:**

$$k^2 + (2k+1) = (k+1)^2.$$

The equality $k^2 = 1+\cdots+(2k-1)$ is the exact point where the hypothesis enters; then the display reaches the $k+1$ formula.

Answer. Start at $1 = 1^2$, assume $P(k)$ for arbitrary $k \geq 1$, and add $2k+1$ to obtain $(k+1)^2$, the required $P(k+1)$.

Frequent A errors

Do not write "assume the claim is true" without naming the case(s). Do not assume $P(k+1)$. For strong induction, do not omit the lower and upper limits of the assumed range. Verify every needed initial case for a multi-step recurrence.

A readiness

Can I state the base, hypothesis, next-case goal, and useful case split without silently assuming the conclusion?

B. Diagnose and Repair an Induction Step

Circular reasoning

An induction step must begin from the induction hypothesis $P(k)$ (or the strong range through k) and derive $P(k+1)$. Beginning with $P(k+1)$ and manipulating it is circular unless every step is a reversible equivalence and the proof is explicitly reorganized into a forward derivation.

A repair workflow

1. Write the actual $P(k)$ and $P(k+1)$ side by side.
2. Delete any use of the target $P(k+1)$ as a premise.
3. Start with the left side or defining object in $P(k+1)$.
4. Isolate a copy of the k-case and cite the IH exactly there.
5. Finish the algebra or inequality, then state the conclusion.

B.1: repair the exponential inequality

Problem (Core B.1). A student tries to prove $2^n \geq n+1$ for every integer $n \geq 0$. In the induction step, the student assumes $2^{k+1} \geq k+2$ and then declares $P(k+1)$ proved. Identify the error, state the correct induction hypothesis, and give a valid chain of inequalities that proves the next case.
Worked solution. The faulty assumption is exactly the target, so the reasoning is circular. **Step 1: delete that assumption.**
Step 2: use the correct hypothesis $2^k \geq k+1$. **Step 3: start from the next-case expression and move forward:**

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k \\ &\geq 2(k+1) && \text{by the IH} \\ &\geq k+2 && \text{because } k \geq 0. \end{aligned}$$

Step 4: close. Thus $P(k+1)$ follows. Notice that the IH is used once and its direction is preserved when multiplying by the positive number 2.
Answer. Assume $2^k \geq k+1$ for arbitrary integer $k \geq 0$, not the target. Then $2^{k+1} \geq 2k+2 \geq k+2$.

Other structural failures

- **Missing base case**: the step alone starts no chain.
- **Too few bases**: a recurrence using two earlier terms needs enough initial cases.
- **Wrong next term**: the added term must be computed at $k+1$.
- **Unjustified strengthening**: proving a stronger statement is fine, but it needs its own base and step.
- **Domain drift**: divisibility, inequalities, or recurrences may require a restricted integer domain.

Repair a wrong next term

Problem (similar example). A student proving $P(n)$: $1+3+\cdots+(2n-1) = n^2$ for integers $n \geq 1$ assumes $P(k)$ and then adds $2k-1$ to the k-term sum to form the $(k+1)$-term sum. Identify and repair the incorrect next term.
Worked solution. After assuming $P(k)$, the next term is $2(k+1)-1 = 2k+1$, not $2k-1$. Hence

$$1+3+\cdots+(2k-1)+(2k+1) = k^2+2k+1 = (k+1)^2.$$

The index, not visual pattern-matching, determines the new term.
Answer. Add $2k+1$, not $2k-1$; the next sum equals $(k+1)^2$.

Two-step recurrences need two bases

Problem (similar example). A sequence is defined by initial values a_0, a_1 and $a_n = a_{n-1} + a_{n-2}$ for every integer $n \geq 2$. A proof of a property $P(n)$ uses both $P(n-1)$ and $P(n-2)$ to establish $P(n)$. The student checks only $P(0)$. Explain the missing base-case check and why it is needed.
Worked solution. When a property of a_n is proved using the two previous cases, verify both initial values. In a strong-induction version, the hypothesis must include enough earlier indices for the recurrence to be defined. One base case cannot support the first step that needs two values.
Answer. Verify both $P(0)$ and $P(1)$; the first recurrence step, at $n = 2$, needs both.

Invalid shortcuts

Do not "cancel" an inequality without checking signs. Do not replace the induction hypothesis with the desired conclusion. Do not test several values and call that an induction proof. A pattern suggests a claim; the proof must cover every allowed integer.

B readiness

Can I point to the first circular or unsupported line, state the correct IH, and replace the broken step with a forward chain that explicitly cites it?

C. The Essential Binomial-Sum Proof

Binomial coefficients and Pascal's identity

$\binom{n}{j}$ counts the j-element subsets of an n-element set. The endpoints satisfy $\binom{n}{0} = \binom{n}{n} = 1$. Pascal's identity is

$$\binom{k+1}{j} = \binom{k}{j} + \binom{k}{j-1} \quad (1 \le j \le k).$$

On the Core, these identities are supplied. Your job is to use them with correct endpoints and indexing.

How to manage the endpoints

For the $(k+1)$-row, separate $j = 0$ and $j = k+1$, then apply Pascal's identity only to $1 \le j \le k$:

$$\sum_{j=0}^{k+1} \binom{k+1}{j} = 1 + \sum_{j=1}^{k} \left[\binom{k}{j} + \binom{k}{j-1} \right] + 1.$$

Reindex the second interior sum with $i = j - 1$; its range becomes $0 \le i \le k-1$.

C.1 complete proof: statement and base

Problem (Core C.1). Prove by mathematical induction that $\sum_{j=0}^{n} \binom{n}{j} = 2^n$ for every integer $n \ge 0$. You may use Pascal's identity $\binom{k+1}{j} = \binom{k}{j} + \binom{k}{j-1}$ for $1 \le j \le k$, and the endpoint identities $\binom{n}{0} = \binom{n}{n} = 1$. Include $P(n)$, the base case, the induction hypothesis, the complete next-case calculation (including reindexing), and the conclusion.

Worked solution, part 1. Let

$$P(n): \quad \sum_{j=0}^{n} \binom{n}{j} = 2^n, \qquad n \ge 0.$$

Base case. At $n = 0$,

$$\sum_{j=0}^{0} \binom{0}{j} = \binom{0}{0} = 1 = 2^0,$$

so $P(0)$ holds.

Induction hypothesis. Fix arbitrary $k \ge 0$ and assume

$$\sum_{j=0}^{k} \binom{k}{j} = 2^k.$$

We must prove the corresponding identity for $k+1$.

C.1 worked solution, continued: induction step

For $k = 0$, the interior sums below are empty and equal 0; both rows sum to 2. For $k \ge 1$, the displayed endpoint and reindexing calculation applies. **Step 1: split endpoints and use Pascal.**

$$\begin{aligned}
\sum_{j=0}^{k+1} \binom{k+1}{j} &= 1 + \sum_{j=1}^{k} \left(\binom{k}{j} + \binom{k}{j-1} \right) + 1 \\
&= 1 + \sum_{j=1}^{k} \binom{k}{j} + \sum_{i=0}^{k-1} \binom{k}{i} + 1 \\
&= \sum_{j=0}^{k} \binom{k}{j} + \sum_{i=0}^{k} \binom{k}{i} \\
&= 2^k + 2^k \qquad \text{by the IH} \\
&= 2^{k+1}.
\end{aligned}$$

The second line uses $i = j - 1$ in the second interior sum. The third line restores the missing endpoint in each copy, so the induction hypothesis can be used on the fourth line. Thus $P(k+1)$ holds. By mathematical induction, $\sum_{j=0}^{n} \binom{n}{j} = 2^n$ for every integer $n \ge 0$.

Conclusion. The identity holds for every integer $n \ge 0$; the two uses of the IH are the two complete k-row sums.

Why the reindexing is valid

When $i = j - 1$, the first old index $j = 1$ becomes $i = 0$, and the last old index $j = k$ becomes $i = k - 1$. The letter itself is a dummy variable; the limits and summand must change together. Adding the two endpoint 1s restores the missing $j = 0$ and $j = k$ terms so that two complete k-row sums appear.

Frequent C errors

Do not apply Pascal's identity at endpoints where the supplied version does not apply. Do not change j to i without changing both limits. Do not cite the IH until the expression is a complete copy of the k-case sum. Do not omit the final induction conclusion.

C readiness

Can I reproduce the proposition, base, IH, endpoint split, reindexing, two complete sums, explicit IH use, and final conclusion without a missing term?

More Worked Practice and Whole-Core Readiness

Divisibility by induction

Problem (similar example). Prove by mathematical induction that $3 \mid (4^n - 1)$ for every integer $n \geq 0$. Use the definition of divisibility: $3 \mid x$ means $x = 3q$ for some integer q.

Worked solution.

Base. $4^0 - 1 = 0$, divisible by 3.

IH. Assume $4^k - 1 = 3q$ for some integer q. Then

$$4^{k+1} - 1 = 4(4^k - 1) + 3 = 4(3q) + 3 = 3(4q + 1).$$

Because $4q + 1$ is an integer, the next expression is divisible by 3. This example shows how a definition supplies the usable form of the IH.

Conclusion. By induction, $3 \mid (4^n - 1)$ for all integers $n \geq 0$.

A finite-sum proof

Problem (similar example). Prove by mathematical induction that $1 + 2 + \cdots + n = n(n+1)/2$ for every integer $n \geq 1$.

Worked solution.

Base: $1 = 1(2)/2$. Assume $1 + \cdots + k = k(k+1)/2$. Then

$$\begin{aligned} 1 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

This is the formula with $n = k + 1$, so induction completes the proof.

Conclusion. The stated sum formula holds for every integer $n \geq 1$.

An inequality with a domain check

Problem (similar example). Prove by mathematical induction that $n! \geq 2^{n-1}$ for every integer $n \geq 1$. Explain where the lower bound on the integer index is used in the induction step.

Worked solution. The base is $1! = 1 = 2^0$. Assuming $k! \geq 2^{k-1}$ for $k \geq 1$,

$$(k+1)! = (k+1)k! \geq 2 \cdot 2^{k-1} = 2^k,$$

because $k + 1 \geq 2$. The domain supplies the comparison needed after the IH.

Conclusion. By induction, the inequality holds for all integers $n \geq 1$. The condition $k \geq 1$ ensures that multiplying $k!$ by $k + 1$ increases it by a factor of at least 2.

What the compact Core certifies

The live Core deliberately has only three problems:

- A: a short induction or strong-induction setup;
- B: diagnosis and repair of a faulty step;
- C: one complete binomial-sum proof.

C is essential. Meets requires a fully correct C and one fully correct short problem. Exceeds requires all three. A retake is a new complete form.

Twenty-minute strategy

1. Read all three prompts and write the exact $P(n)$ for C.
2. Complete A or B cleanly in about 3–4 minutes.
3. Reserve 12–13 minutes for C.
4. Use the last minutes to check base values, index limits, where the IH was cited, and the final conclusion.

Do not sacrifice the essential C proof to over-polish a short response.

Proof language that matters

"Assume true for n" is ambiguous; name the arbitrary index and exact hypothesis. "Therefore true" is incomplete; state that $P(k+1)$ follows and invoke induction over the stated domain. Algebra alone does not replace the logical structure.

Whole-Core readiness

I can:

- distinguish ordinary from strong induction;
- state every component of a valid plan;
- find and repair circular use of the target;
- calculate the correct next term;
- separate endpoints and reindex a binomial sum;
- show exactly where the induction hypothesis is used; and
- finish with a correctly scoped induction conclusion.